DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -22 DEGREES FAHRENHEIT (-30 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 1,000 VOLTS, TOPSIDE, MULTI-CONDUCTOR CIRCUIT INTEGRITY TYPES LSE1CI, LSE2CI, LSE3CI, LSE4CI, LSE6CI, LSE7CI, LSE10CI, LSE14CI, LSE19CI, LSE30CI, LSE37CI, LSE44CI, LSE61CI, AND LSE91CI (INCLUDING VARIANTS LSE1CIW, LSE2CIW, LSE3CIW, LSE4CIW, LSE6CIW, LSE7CIW, LSE10CIW, LSE14CIW, LSE19CIW, LSE30CIW, LSE37CIW, LSE44CIW, LSE61CIW, AND LSE91CIW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LSE**CI (non-watertight with circuit integrity), and LSE**CIW (watertight with circuit integrity).

Construction (Non-Watertight and Watertight) with Circuit Integrity

First	-	Copper conductor, coated or uncoated, ASTM B8 or Navy Standard (see table I or II for size and number of strands).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation, extruded or taped (see table I or II for thickness).
Fourth	-	An optional glass braid.
Fifth	-	An optional covering unless required (see 3.3.6 of MIL-DTL-24643).
Sixth	-	The required number of conductors (see table I or II) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded cable assembly. (Single conductor cables are exempt from the cabling requirements stated above.) Conductor identification codes applied using method 1 or 3, or letter identification code applied by method 5. If method 1 is used, outer surface to be printed shall appear black. Single conductor cables shall be colored white or natural with no other identification applied. Cables with more than four conductors may not employ the letter identification code.
Seventh	-	An optional binder.
Eighth	-	Copper foil/polyester tape. Applied helically with 50 percent minimum overlap (metallic foil side facing out and in contact with the braid).
Ninth	-	Braided shield of tin-coated copper. Braid must be constructed such that the completed cable complies with the surface transfer impedance requirements specified herein. In addition, the minimum coverage of the braid shall be 90 percent and the minimum shield strand size shall be 34 AWG.

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Tenth - Copper foil/polyester tape/copper foil. Applied helically with 50 percent minimum overlap. Tape must provide electrical contact between the inner and outer braided shields.

Eleventh - Braided shield of tin-coated copper. Braid must be constructed such that the completed cable complies with the surface transfer impedance requirements specified herein. In addition, the minimum coverage of the braid shall be 90 percent and the minimum shield strand size shall be 34 AWG.

For watertight constructions only, the use of solid or semi-solid water-block materials (such as silicone, hot melts, or gels) is not allowed on the outer braid. Watertightness shall be by means that could include, but are not limited to, water-swell yarns incorporated within or over the braid or water-swell tape helically applied over the outer braid with 25 percent minimum overlap. The water-block material shall not stick or adhere to the outer braid strands.

Twelfth - An optional binder (tape must not be clear).

Thirteenth - Cross-linked polyolefin jacket. See table I or II for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details (non-watertight).

Military part no. M24643/85	Type and size	Number of conductors	Conductor size (AWG/ MCM)	Number of strands	Insulation thickness min. avg. (inch)	Jacket thickness min. avg. (inch)	Cable diameter (inches)		Conductor resistance (ohms)	Accelerated service loading (amperes)
							min.	max.		
-01UO	LSE1CI-2	1	18	7	0.018	0.045	0.290	0.335	6.93	None
-02UO	LSE2CI-2	2	18	7	0.018	0.045	0.390	0.440	7.07	None
-03UO	LSE3CI-2	3	18	7	0.018	0.045	0.405	0.455	7.07	None
-04UO	LSE4CI-2	4	18	7	0.018	0.045	0.430	0.480	7.07	None
-05UO	LSE7CI-2	7	18	7	0.018	0.045	0.490	0.540	7.07	None
-06UO	LSE10CI-2	10	18	7	0.018	0.050	0.600	0.660	7.07	None
-07UO	LSE14CI-2	14	18	7	0.018	0.050	0.640	0.700	7.07	None
-08UO	LSE19CI-2	19	18	7	0.018	0.055	0.710	0.770	7.07	None
-09UO	LSE30CI-2	30	18	7	0.018	0.060	0.860	0.930	7.07	None
-10UO	LSE37CI-2	37	18	7	0.018	0.060	0.920	0.990	7.07	None
-11UO	LSE44CI-2	44	18	7	0.018	0.060	1.015	1.095	7.07	None
-12UO	LSE61CI-2	61	18	7	0.018	0.065	1.125	1.210	7.07	None
-13UO	LSE91CI-2	91	18	7	0.018	0.065	1.330	1.415	7.07	None
-14UO	LSE2CI-3	2	16	7	0.018	0.045	0.410	0.460	4.30	None
-15UO	LSE3CI-3	3	16	7	0.018	0.045	0.425	0.480	4.30	None
-16UO	LSE4CI-3	4	16	7	0.018	0.045	0.455	0.510	4.30	None
-17UO	LSE7CI-3	7	16	7	0.018	0.045	0.525	0.575	4.30	None
-18UO	LSE2CI-4	2	14	7	0.018	0.045	0.440	0.490	2.68	None
-19UO	LSE3CI-4	3	14	7	0.018	0.045	0.455	0.510	2.68	None
-20UO	LSE4CI-4	4	14	7	0.018	0.045	0.490	0.540	2.68	None
-21UO	LSE7CI-4	7	14	7	0.018	0.045	0.565	0.615	2.68	None
-22UO	LSE2CI-9	2	10	7	0.018	0.045	0.520	0.570	1.08	None
-23UO	LSE3CI-9	3	10	7	0.018	0.045	0.545	0.595	1.08	None
-24UO	LSE4CI-9	4	10	7	0.018	0.050	0.600	0.665	1.08	None
-25UO	LSE2CI-14	2	9	7	0.018	0.045	0.560	0.615	0.859	100
-26UO	LSE3CI-14	3	9	7	0.018	0.050	0.595	0.655	0.859	90
-27UO	LSE1CI-23	1	7	7	0.018	0.045	0.405	0.455	0.534	140
-28UO	LSE2CI-23	2	7	7	0.018	0.050	0.630	0.690	0.543	130
-29UO	LSE3CI-23	3	7	7	0.018	0.050	0.660	0.725	0.543	115
-30UO	LSE4CI-23	4	7	7	0.018	0.055	0.735	0.805	0.543	115
-31UO	LSE1CI-50	1	3	19	0.035	0.045	0.540	0.590	0.209	240
-32UO	LSE2CI-50	2	3	19	0.035	0.060	0.915	0.985	0.210	215

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TABLE I. Details (non-watertight) – Continued.

Military part no. M24643/85	Type and size	Number of conductors	Conductor size (AWG/ MCM)	Number of strands	Insulation thickness min. avg. (inch)	Jacket thickness min. avg. (inch)	Cable diameter (inches)		Conductor resistance (ohms)	Accelerated service loading (amperes)
							min.	max.		
-33UO	LSE2CI-50	3	3	19	0.035	0.060	0.960	1.045	0.210	185
-34UO	LSE4CI-50	4	3	19	0.035	0.060	1.070	1.150	0.210	185
-35UO	LSE2CI-60	2	2	37	0.035	0.060	0.980	1.060	0.190	235
-36UO	LSE3CI-60	3	2	37	0.035	0.060	1.040	1.120	0.190	215
-37UO	LSE1CI-75	1	1	37	0.035	0.050	0.615	0.675	0.132	315
-38UO	LSE2CI-75	2	1	37	0.035	0.060	1.050	1.130	0.134	285
-39UO	LSE3CI-75	3	1	37	0.035	0.065	1.125	1.205	0.134	250
-40UO	LSE4CI-75	4	1	37	0.035	0.065	1.250	1.330	0.134	250
-41UO	LSE1CI-100	1	0	61	0.035	0.050	0.655	0.715	0.105	375
-42UO	LSE2CI-100	2	0	61	0.035	0.065	1.140	1.220	0.106	335
-43UO	LSE3CI-100	3	0	61	0.035	0.065	1.210	1.290	0.106	295
-44UO	LSE4CI-100	4	0	61	0.035	0.065	1.350	1.430	0.106	295
-45UO	LSE6CI-100	6	0	61	0.035	0.065	1.605	1.685	0.106	475
-46UO	LSE1CI-125	1	00	61	0.040	0.055	0.725	0.795	0.0875	415
-47UO	LSE2CI-125	2	00	61	0.040	0.065	1.270	1.350	0.0888	380
-48UO	LSE3CI-125	3	00	61	0.040	0.065	1.350	1.430	0.0888	340
-49UO	LSE1CI-150	1	000	61	0.040	0.055	0.770	0.840	0.0663	490
-50UO	LSE2CI-150	2	000	61	0.040	0.065	1.360	1.440	0.0674	445
-51UO	LSE3CI-150	3	000	61	0.040	0.065	1.445	1.525	0.0674	400
-52UO	LSE4CI-150	4	000	61	0.040	0.075	1.640	1.720	0.0674	400
-53UO	LSE6CI-150	6	000	61	0.040	0.075	1.955	2.035	0.0674	570
-54UO	LSE1CI-200	1	0000	61	0.050	0.060	0.865	0.935	0.0525	580
-55UO	LSE2CI-200	2	0000	61	0.050	0.065	1.530	1.610	0.0530	525
-56UO	LSE3CI-200	3	0000	61	0.050	0.075	1.650	1.730	0.0530	460
-57UO	LSE4CI-200	4	0000	61	0.050	0.075	1.845	1.930	0.0530	465
-58UO	LSE6CI-200	6	0000	61	0.050	0.075	2.210	2.290	0.0530	650
-59UO	LSE1CI-300	1	300	91	0.050	0.060	0.955	1.030	0.0370	750
-60UO	LSE2CI-300	2	300	91	0.050	0.075	1.735	1.820	0.0377	700
-61UO	LSE3CI-300	3	300	91	0.050	0.075	1.850	1.935	0.0377	590
-62UO	LSE1CI-400	1	400 (Navy STD)	127	0.050	0.060	1.065	1.150	0.0268	925
-63UO	LSE2CI-400	2	400 (Navy STD)	127	0.050	0.075	1.965	2.050	0.0273	835
-64UO	LSE3CI-400	3	400 (Navy STD)	127	0.050	0.075	2.100	2.180	0.0273	750

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TABLE II. Details (watertight).

Military part no. M24643/85	Type and size	Number of conductors	Conductor size (AWG/ MCM)	Number of strands	Insulation thickness min. avg. (inch)	Jacket thickness min. avg. (inch)	Cable diameter (inches)		Conductor resistance (ohms)	Accelerated service loading (amperes)
							min.	max.		
-65UO	LSE1CIW-2	1	18	7	0.018	0.045	0.319	0.359	6.87	None
-66UO	LSE2CIW-2	2	18	7	0.018	0.045	0.414	0.464	7.07	None
-67UO	LSE3CIW-2	3	18	7	0.018	0.045	0.429	0.479	7.07	None
-68UO	LSE4CIW-2	4	18	7	0.018	0.045	0.455	0.505	7.07	None
-69UO	LSE7CIW-2	7	18	7	0.018	0.045	0.514	0.564	7.07	None
-70UO	LSE10CIW-2	10	18	7	0.018	0.050	0.619	0.679	7.07	None
-71UO	LSE14CIW-2	14	18	7	0.018	0.050	0.660	0.720	7.07	None
-72UO	LSE19CIW-2	19	18	7	0.018	0.055	0.724	0.794	7.07	None
-73UO	LSE30CIW-2	30	18	7	0.018	0.060	0.875	0.945	7.07	None
-74UO	LSE37CIW-2	37	18	7	0.018	0.060	0.934	1.004	7.07	None
-75UO	LSE44CIW-2	44	18	7	0.018	0.060	1.029	1.109	7.07	None
-76UO	LSE61CIW-2	61	18	7	0.018	0.065	1.139	1.219	7.07	None
-77UO	LSE91CIW-2	91	18	7	0.018	0.065	1.339	1.419	7.07	None
-78UO	LSE2CIW-3	2	16	7	0.018	0.045	0.442	0.492	4.30	None
-79UO	LSE3CIW-3	3	16	7	0.018	0.045	0.459	0.509	4.30	None
-80UO	LSE4CIW-3	4	16	7	0.018	0.045	0.489	0.539	4.30	None
-81UO	LSE7CIW-3	7	16	7	0.018	0.045	0.556	0.606	4.30	None
-82UO	LSE2CIW-4	2	14	7	0.018	0.045	0.472	0.522	2.68	None
-83UO	LSE3CIW-4	3	14	7	0.018	0.045	0.491	0.541	2.68	None
-84UO	LSE4CIW-4	4	14	7	0.018	0.045	0.525	0.575	2.68	None
-85UO	LSE7CIW-4	7	14	7	0.018	0.050	0.606	0.666	2.68	None
-86UO	LSE2CIW-9	2	10	7	0.018	0.045	0.558	0.608	1.08	None
-87UO	LSE3CIW-9	3	10	7	0.018	0.045	0.579	0.639	1.08	None
-88UO	LSE4CIW-9	4	10	7	0.018	0.050	0.640	0.700	1.08	None
-89UO	LSE2CIW-14	2	9	7	0.018	0.050	0.595	0.660	0.859	100
-90UO	LSE3CIW-14	3	9	7	0.018	0.050	0.625	0.685	0.859	90
-91UO	LSE1CIW-23	1	7	7	0.018	0.045	0.430	0.485	0.534	140
-92UO	LSE2CIW-23	2	7	7	0.018	0.050	0.660	0.720	0.543	130
-93UO	LSE3CIW-23	3	7	7	0.018	0.055	0.700	0.765	0.543	115
-94UO	LSE4CIW-23	4	7	7	0.018	0.055	0.760	0.835	0.543	115
-95UO	LSE1CIW-50	1	3	19	0.035	0.045	0.565	0.620	0.209	240
-96UO	LSE2CIW-50	2	3	19	0.035	0.060	0.940	1.015	0.210	215
-97UO	LSE2CIW-50	3	3	19	0.035	0.060	0.990	1.075	0.210	185
-98UO	LSE4CIW-50	4	3	19	0.035	0.060	1.100	1.180	0.210	185
-99UO	LSE2CIW-60	2	2	37	0.035	0.060	1.010	1.090	0.190	240
-100UO	LSE3CIW-60	3	2	37	0.035	0.060	1.065	1.150	0.190	215
-101UO	LSE1CIW-75	1	1	37	0.035	0.050	0.645	0.705	0.132	315
-102UO	LSE2CIW-75	2	1	37	0.035	0.060	1.080	1.160	0.134	285
-103UO	LSE3CIW-75	3	1	37	0.035	0.065	1.155	1.235	0.134	250
-104UO	LSE4CIW-75	4	1	37	0.035	0.065	1.280	1.360	0.134	250
-105UO	LSE1CIW-100	1	0	61	0.035	0.050	0.685	0.745	0.105	375
-106UO	LSE2CIW-100	2	0	61	0.035	0.065	1.170	1.250	0.106	335
-107UO	LSE3CIW-100	3	0	61	0.035	0.065	1.240	1.320	0.106	295
-108UO	LSE4CIW-100	4	0	61	0.035	0.065	1.380	1.460	0.106	295
-109UO	LSE6CIW-100	6	0	61	0.035	0.075	1.655	1.735	0.106	475
-110UO	LSE1CIW-125	1	00	61	0.040	0.055	0.755	0.825	0.0875	415
-111UO	LSE2CIW-125	2	00	61	0.040	0.065	1.300	1.380	0.0888	380
-112UO	LSE3CIW-125	3	00	61	0.040	0.065	1.380	1.460	0.0888	340
-113UO	LSE1CIW-150	1	000	61	0.040	0.055	0.800	0.870	0.0663	490
-114UO	LSE2CIW-150	2	000	61	0.040	0.065	1.390	1.470	0.0674	445
-115UO	LSE3CIW-150	3	000	61	0.040	0.065	1.475	1.555	0.0674	400

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TABLE II. Details (watertight) – Continued.

Military part no. M24643/85	Type and size	Number of conductors	Conductor size (AWG/ MCM)	Number of strands	Insulation thickness min. avg. (inch)	Jacket thickness min. avg. (inch)	Cable diameter (inches)		Conductor resistance (ohms)	Accelerated service loading (amperes)
							min.	max.		
-116UO	LSE4CIW-150	4	000	61	0.040	0.075	1.670	1.750	0.0674	400
-117UO	LSE6CIW-150	6	000	61	0.040	0.075	1.985	2.065	0.0674	570
-118UO	LSE1CIW-200	1	0000	61	0.050	0.060	0.895	0.965	0.0525	580
-119UO	LSE2CIW-200	2	0000	61	0.050	0.065	1.560	1.640	0.0530	525
-120UO	LSE3CIW-200	3	0000	61	0.050	0.075	1.675	1.760	0.0530	460
-121UO	LSE4CIW-200	4	0000	61	0.050	0.075	1.875	1.955	0.0530	465
-122UO	LSE6CIW-200	6	0000	61	0.050	0.075	2.240	2.320	0.0530	650
-123UO	LSE1CIW-300	1	300	91	0.050	0.060	0.985	1.065	0.0370	750
-124UO	LSE2CIW-300	2	300	91	0.050	0.065	1.770	1.850	0.0377	700
-125UO	LSE3CIW-300	3	300	91	0.050	0.075	1.880	1.965	0.0377	590
-126UO	LSE1CIW-400	1	400 (Navy STD)	127	0.050	0.060	1.1.00	1.180	0.0268	925
-127UO	LSE2CIW-400	2	400 (Navy STD)	127	0.050	0.075	2.000	2.100	0.0273	835
-128UO	LSE3CIW-400	3	400 (Navy STD)	127	0.050	0.075	2.130	2.230	0.0273	750

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.) — See table I or II

Voltage withstand (volts, root mean square, min.)

Conductor to conductor (multi-conductor cables only)

Sizes 2 through 9 — 3,000

Sizes 14 and larger — 5,000

Conductor to shield

Sizes 2 through 9 — 1,500

Sizes 14 and larger — 2,500

Insulation resistance (megohms/1,000 feet, min.) — Not applicable

Conductor to conductor (multi-conductor cables only)

Sizes 2 through 23 — 500

Size 50 — 200

Sizes 60 through 125 — 150

Sizes 150 and larger — 125

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Conductor to shield	
Sizes 2 through 23	250
Size 50	100
Sizes 60 through 125	75
Sizes 150 and larger	60
Conductor and shield continuity	No failure
Jacket flaws	No failure
Group A:	
Visual and dimensional	No failure
Watertightness (applicable to watertight constructions only) (see MIL-DTL-24643 for limits of water leakage)	No failure
Abrasion resistance (insulation) (scrapes, min.) (applicable to constructions using glass braids with coverings)	250
Crack resistance (insulation) (applicable to constructions using glass braids with coverings)	No damage
Group B:	
Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation (extruded insulation only)	50
Jacket (when tested at 392 °F [200 °C])	0
Drip (203±1.8 °F [95±1 °C]) (applicable to watertight constructions only) (see MIL-DTL-24643 for limits of drip allowance)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

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Group C:

Physicals (aged) (air oven)	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation) (cycles, min.) (applicability: cables using method 1 or method 3)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability (applicable to watertight constructions only)	No failure
Shrinkage	No failure
Shield (conformance to material, construction, and coverage)	No failure
Surface transfer impedance (outer shields only)	
Milliohms per meter (max.)	5.0
Electromagnetic pulse (EMP) response (dB, min.)	80

Group D:

Gas flame (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (-22±3.6 °F [-30±2 °C])	No damage
Gas flame (circuit integrity) (3 hours)	No failure
Accelerated service (see table I or II for applicability and load current)	No failure
Aging and compatibility (cable) (203±3.6 °F [95±2 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	

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Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption (extruded insulation only)	No failure
Long term jacket immersion	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSE1CI through LSE4CI	1,000
LSE6CI through LSE19CI	800
LSE30CI through LSE91CI	500

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2017-006)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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